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TREND-BASED MOCK PAPER

GATE Geology and Geophysics

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Geology and Geophysics
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Choose similar to 'volcanic':

(Numerical Answer Type — Enter the numerical value)

Section B: Geology and Geophysics

Q2. [1-MCQ] [1 mark]

Mean of 1,2,3,4,5:

- (A) 2
 - (B) 3
 - (C) 4
 - (D) 5
-

Q3. [1-NAT] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
 - (B) 6
 - (C) 36
 - (D) 12
-

Q5. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$

- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q6. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) ∞

Q7. [1-NAT] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q9. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q10. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q11. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q12. [1-MCQ] [1 mark]

$\frac{d}{dx} [\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q13. [1-MCQ] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q14. [1-NAT] [1 mark]

The Laplace transform of $t \cdot e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q15. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q16. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q17. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q18. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q19. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q20. [1-MCQ] [1 mark]

The rank of the matrix $[[1,2,3],[2,4,6],[3,6,9]]$ is:

- (A) 1
- (B) 2
- (C) 3

(D) 0

Q21. [2-NAT] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q23. [2-NAT] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q25. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q26. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1,0]^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q28. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q29. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q30. [2-MCQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q31. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q32. [2-NAT] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q33. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a, b]$ and differentiable on (a, b)
- (D) Both continuous and differentiable everywhere

Q34. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q36. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MSQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q39. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q40. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)} \cdot u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q41. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q42. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q43. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q44. [2-MCQ] [2 marks]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q45. [1-MCQ] [1 mark]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q46. [1-MCQ] [1 mark]

Study of fossils:

- (A) Paleontology
- (B) Petrology
- (C) Mineralogy
- (D) None

Q47. [1-NAT] [1 mark]

Law of superposition by:

(Numerical Answer Type — Enter the numerical value)

Q48. [1-MCQ] [1 mark]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q49. [1-MCQ] [1 mark]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q50. [2-NAT] [2 marks]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q51. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q52. [2-MSQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q53. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q54. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q55. [2-NAT] [2 marks]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q56. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q57. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q58. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q59. [2-NAT] [2 marks]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q60. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q61. [1-NAT] [1 mark]

Earth's magnetic field generated by:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

P-waves are:

- (A) Primary
 - (B) Secondary
 - (C) Surface
 - (D) Love
-

Q63. [1-MCQ] [1 mark]

Gravity anomaly due to:

- (A) Density variation
 - (B) Magnetism
 - (C) Electricity
 - (D) None
-

Q64. [1-MCQ] [1 mark]

Landsat uses:

- (A) Optical
 - (B) Radar
 - (C) Both
 - (D) None
-

Q65. [1-NAT] [1 mark]

Igneous rock from:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1NAT	A	1
6	1NAT	A	1
7	1MCQ	A	1
8	1MCQ	A	1
9	1MCQ	A	1
10	1NAT	B	1
11	1MCQ	B	1
12	1NAT	A	1
13	1MCQ	A	1

14	1MCQ	B	1
15	1MCQ	B	1
16	1MCQ	B	1
17	1NAT	A	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1MCQ	A	1
21	1MCQ	B	1
22	1MCQ	C	1
23	1MCQ	B	1
24	1NAT	A	1
25	1MCQ	C	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1MCQ	C	1

29	1MCQ	C	1
30	1MCQ	A	1
31	2NAT	B	2
32	2MCQ	A	2
33	2NAT	A	2
34	2MCQ	A	2
35	2MCQ	B	2
36	2NAT	A	2
37	2MCQ	C	2
38	2NAT	B	2
39	2MCQ	B	2
40	2MCQ	B	2
41	2MCQ	A	2
42	2MCQ	A	2
43	2MSQ	B	2

44	2MCQ	A	2
45	2MCQ	B	2
46	2NAT	B	2
47	2MCQ	C	2
48	2NAT	A	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2NAT	B	2
52	2NAT	C	2
53	2NAT	A	2
54	2MCQ	B	2
55	2MCQ	B	2
56	2MSQ	A	2
57	2MCQ	B	2
58	2MCQ	B	2

59	2NAT	B	2
60	2MCQ	A	2
61	2MCQ	C	2
62	2MCQ	B	2
63	2MCQ	A	2
64	2MCQ	B	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Statistics | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Geology | Topic: Minerals | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Geology | Topic: Fossils | Difficulty: difficult

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Geology | Topic: Stratigraphy | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Magnetism | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Seismic | Difficulty: easy

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Gravity | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Remote Sensing & GIS | Topic: Satellite | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Petrology | Topic: Rocks | Difficulty: moderate

Q11.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q12.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q13.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q14.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q15.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q16.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q17.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q21.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q25.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q29.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q30.

Answer: A

Solution:

Row 2 = $2 \times$ Row 1, Row 3 = $3 \times$ Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q39.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: difficult

Q41.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

$L\{tf(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q44.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q46.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q48.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q51.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: difficult

Q52.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q54.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q55.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: difficult

Q56.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q58.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q59.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q60.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q64.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q65.

Answer: A

Solution:

$L\{t \cdot f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 43 questions, 67 marks
Geology: 16 questions, 27 marks
Geophysics: 3 questions, 3 marks
General Aptitude: 1 questions, 1 marks
Remote Sensing & GIS: 1 questions, 1 marks
Petrology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 20 | Moderate: 37 | Difficult: 8

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.